

W1

Algebra For Honor Math

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Contents

Q3

(a)

$$\forall m \in \mathbb{R}, \sin^3(2m + 1) = \frac{1}{2}.$$

(b)

$$\exists \theta \in \mathbb{R}, \forall \phi \in \mathbb{R}, \sin(2\theta) < \cos(\phi - \theta).$$

Q4

(a)

$$\forall x \in \mathbb{R}, f(x) \leq 3.$$

(b)

$$\forall x \in \mathbb{R}, \exists a, b \in \mathbb{R}, x = \frac{a + b}{2}$$

Q5

- (a) This is a true statement.
- (b) This is a false statement.
- (c) This is an open sentence which relies on q .
- (d) This is a true statement.
- (e) This is a true statement.

Q6

- (a) $S = \mathcal{T}, P(x, y) = P_2(x, y).$
- (b) $S = \mathbb{N}, P(x, y) = P_3(x, y).$
- (c) $S = \mathbb{Z}, P(x, y) = P_4(x, y).$
- (d) $S = \mathbb{Q}, P(x, y) = P_1(x, y).$